Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team OpsCrew

Expense exception plans with human confirmation

Captain: Sunday Lee
Theme: Future of Work & Enterprise Productivity

Brief about the idea

OpsCrew helps operations staff turn a synthetic expense exception into a supported handling plan.

Gemini extracts facts, proposes steps, then reviews the proposal in a separate call. Fixed policy checks constrain each stage.

A person reviews the current plan and explicitly confirms a case. The app reads the saved record independently before showing success.

The prototype covers one synthetic USD policy. It creates handling records and performs no reimbursement or payment.

Opportunities and proposed value

Problem: staff must reconcile intake text, missing receipts and policy thresholds before deciding what to request next.

Proposed difference: OpsCrew binds the plan to supported facts and exact policy clauses, then stores the person's confirmation with that version.

Design value: input edits invalidate old plans. Contradictions and unsupported citations block case creation.

Expected impact: clearer exception triage and fewer unsupported actions. Actual staff time savings and customer adoption remain unmeasured.

Features offered by the solution

Synthetic presets and original text citations make intake decisions inspectable.

Facts, Planner and Reviewer outputs expose disagreements and exact policy evidence.

Firebase visitor identity, owned history and version-bound confirmation protect each person's handling records.

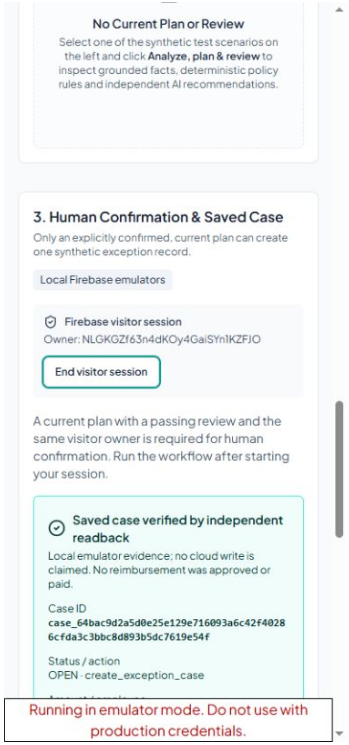
Atomic writes, idempotent recovery and independent readback prevent duplicate or unverified success. Shared quotas bound model traffic.

Process flow and confirmation



Missing evidence produces a request plan. Contradictions block progression.
Editing intake requires a new analysis and a new confirmation.

Interface and narrow-screen review



Observed local browser behavior

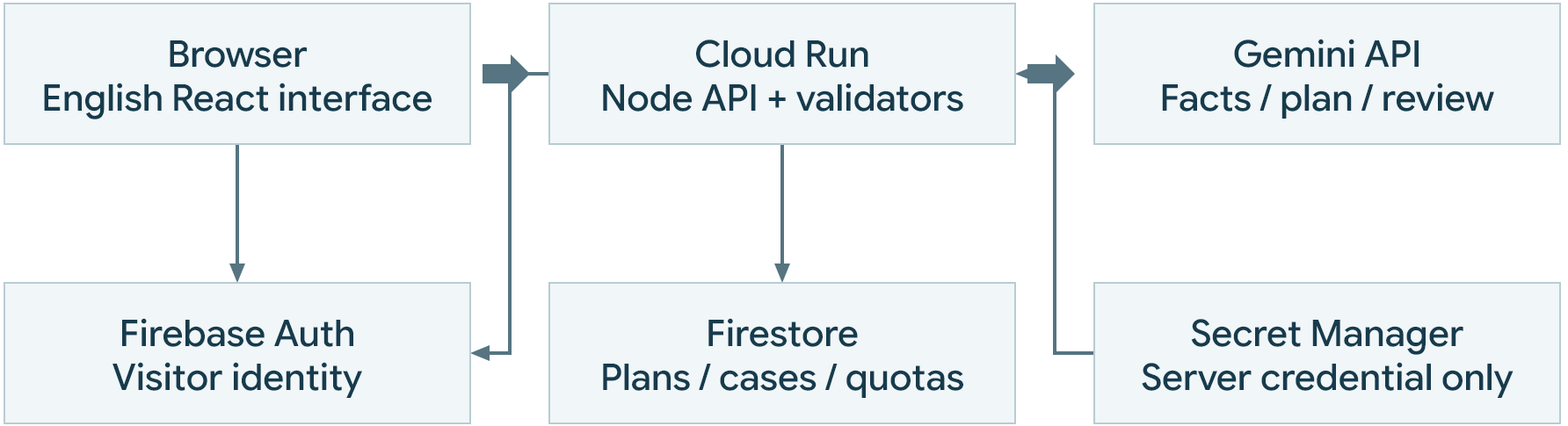
Explicit confirmation remains separate from the AI review.

Saved history survives a page reload with the same visitor identity.

The 390 px viewport had no horizontal overflow. Keyboard focus was visible.

Historical screenshot: local emulators and injected model output, 10 October. The cloud API evidence on pages 10-11 is separate.

Cloud architecture and trust boundaries



Recorded cloud APIs verify owner access, confirmation and saved-case readback. Direct visitor Firestore access is denied. Source-to-image identity and complete resource controls await verification.

Technologies used in the solution

Component	Role
Google AI Studio Build	Fresh application origin on 7 October
Gemini 3.1 Flash-Lite	Facts, separately prompted Planner and Reviewer
Node 24 / React / TypeScript	API, explicit review interface and fixed validators
Firebase Auth + Firestore	Visitor ownership, atomic records and shared quotas
Cloud Run + Secret Manager	Cloud API runtime and server-only credential

Created from zero in AI Studio Build. The teammate used Antigravity to assist Cloud deployment (owner-reported).

Implementation cost and traffic controls

Team management target: USD 40. Billing alerts and application quotas cannot guarantee a monetary spending cap.

Default model limits: 2 concurrent runs globally, 1 per visitor. Each visitor has 2 runs per minute and 12 per UTC day.

Global limits: 6 runs per minute and 60 per UTC day. Each workflow reserves up to 6 provider attempts. Unknown usage receives no refund.

13 October observations: 10 provider attempts and 11,680 known tokens. One first FACTS call returned 503. Its usage and actual billed cost are unknown.

Prototype: recorded cloud API results

Synthetic input / check	Observed cloud result
USD 38, missing receipt	EVIDENCE_REQUEST plan. Explicit confirmation saved one case; independent readback matched.
DEV-01, receipt available	NO_ACTION_REQUIRED. No exception case.
DEV-10, USD 38 form / USD 50 text	BLOCKED. Conflicting sources prevent case creation.
Replay / different visitor	Same committed case / 404 for the other visitor.

Recorded HTTP response evidence, 13 October 2026. Synthetic handling records only. No reimbursement approval or payment occurs.

Performance report and evidence boundaries

Evidence	Result	Method
Cloud core API	11 / 11	Real Auth, Gemini, Firestore; 13 Oct
Cloud safety API	14 / 14	Includes core verifier; overlapping checks
Real Gemini workflows	3 public cases	Missing receipt, no action, contradiction
Offline engineering	232 / 232	2026-10-12 local regressions
Historical packaging	12 groups	Includes 9 HTTP checks; native runtime, 11 Oct

Three cloud workflow observations: 8.381-12.042 s. The missing-receipt run recovered from one 503. These public cases establish neither an SLA nor independent held-out acceptance.

Additional details and future development

Observed cloud APIs complete plan, explicit confirmation and independent record readback. Source-to-image identity and full operational controls still require verification.

The teammate seals eight independent cases before first evaluation. Retain every attempt and run the main demo successfully three times in sequence.

Record a genuine browser flow on the accepted release. Verify the final sub-three-minute English video through signed-out playback, and complete the team's registration declarations.

After evaluation, consider broader policies and staff timing studies. Production reimbursement integrations require a separate design and authorization.

Repository, demo and final product links

Public GitHub repository

github.com/sundaylee3100-ljl/AI-Builder-Cup-2026-OpsCrew

Cloud product URL

https://opscrew-883881551915.us-central1.run.app

Public demo video (Verified final release)

https://youtu.be/8MACVIPWCt8

Final release version. Public links verified.
Demonstration video recorded live against Cloud Run deployment.